

Grade 6 Accelerated
Unit 11
Lesson 1 Practice

Name Answer Key
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Find the sum for each expression in questions 1-6.

1. $(12.53) + (-27.37) = -14.84$

$$\begin{array}{r} 27.37 \\ -12.53 \\ \hline 14.84 \end{array}$$

4. $\frac{14}{3} + 5\frac{3}{7} = 10\frac{2}{21}$

$$4\frac{2}{3} + 5\frac{3}{7} = 4\frac{14}{21} + 5\frac{9}{21}$$

$$9\frac{23}{21} = 10\frac{2}{21}$$

2. $-16.29 + (-24.73) = -41.02$

$$\begin{array}{r} 24.73 \\ +16.29 \\ \hline 41.02 \end{array}$$

5. $(37.29) + (15.36) = 52.65$

$$\begin{array}{r} 37.29 \\ +15.36 \\ \hline 52.65 \end{array}$$

3. $-3\frac{4}{5} + \frac{13}{4} = \frac{-11}{20}$ or -0.55

$$\frac{-19}{5} + \frac{13}{4}$$

$$\frac{-76}{20} + \frac{65}{20}$$

$$\frac{-11}{20}$$

6. $7\frac{5}{6} + \left(-4\frac{6}{8}\right) = 3\frac{1}{12}$

$$7\frac{5}{6} + \left(-4\frac{3}{4}\right)$$

$$7\frac{10}{12} + \left(-4\frac{9}{12}\right)$$

$$7\frac{10}{12} - 4\frac{9}{12}$$

$$3\frac{1}{12}$$

Using the expression bank, find a matching expression for questions 7-10.

Expression Bank

$$\frac{39}{5} + \frac{3}{2} = \frac{78}{10} + \frac{15}{10} = 9\frac{3}{10}$$

$$\left(-13\frac{7}{8}\right) + 24\frac{14}{16} = \left(-13\frac{7}{8}\right) + 24\frac{7}{8} = 11$$

$$\frac{62}{15} + \frac{19}{10} = \frac{124}{30} + \frac{57}{30} = 6\frac{1}{30}$$

$$12\frac{3}{8} + \left(-5\frac{1}{24}\right) = 12\frac{9}{24} + \left(-5\frac{1}{24}\right) = 7\frac{8}{24} = 7\frac{1}{3}$$

7. $\frac{14}{4} + 3\frac{5}{6} = 12\frac{3}{8} + \left(-5\frac{1}{24}\right)$

9. $13\frac{3}{9} + \left(-\frac{14}{6}\right) = \left(-13\frac{7}{8}\right) + 24\frac{14}{16}$

$$3\frac{1}{2} + 3\frac{5}{6}$$

$$13\frac{3}{9} - 2\frac{2}{6}$$

$$3\frac{3}{6} + 3\frac{5}{6}$$

$$13\frac{1}{3} - 2\frac{1}{3}$$

$$6\frac{8}{6} = 7\frac{2}{6} = 7\frac{1}{3}$$

$$11$$

8. $-5\frac{4}{8} + 14\frac{8}{10} = \frac{39}{5} + \frac{3}{2}$

10. $14\frac{10}{12} + \left(-\frac{44}{5}\right) = \frac{62}{15} + \frac{19}{10}$

$$-5\frac{1}{2} + 14\frac{8}{10}$$

$$14\frac{5}{6} - 8\frac{4}{5}$$

$$-5\frac{5}{10} + 14\frac{8}{10}$$

$$14\frac{25}{30} - 8\frac{24}{30}$$

$$9\frac{3}{10}$$

$$6\frac{1}{30}$$

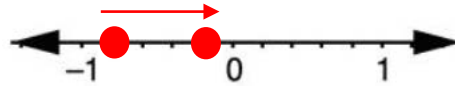
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Lesson 2 Practice

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Find the difference using a number line representation for each phrase for questions 1-2.

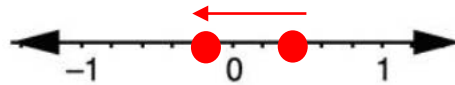
1. The difference of $-\frac{1}{5}$ and $-\frac{4}{5}$.

$$-\frac{1}{5} - \left(-\frac{4}{5}\right) = \frac{3}{5} = 0.6$$



2. The difference of -0.2 and 0.4 .

$$-0.2 - 0.4 = -0.6$$



Find the difference of each expression in questions 3-6.

3. $(-6.79) - (-1.15)$

$$-6.79 + 1.15 = -5.64$$

$$\begin{array}{r} 6.79 \\ -1.15 \\ \hline 5.64 \end{array}$$

5. $\left(\frac{57}{4}\right) - \left(\frac{13}{2}\right)$

$$\frac{57}{4} - \frac{26}{4} = \frac{31}{4} = 7\frac{3}{4}$$

4. $4\frac{5}{6} - 3\frac{6}{9}$

$$4\frac{15}{18} - 3\frac{12}{18} = 1\frac{3}{18} = 1\frac{1}{6}$$

6. $-4.97 - (1.54)$

$$-4.97 + (-1.54) = -6.51$$

$$\begin{array}{r} 4.97 \\ +1.54 \\ \hline 6.51 \end{array}$$

7. What is the difference of -8.71 and 3.02 ?

$$\begin{array}{r} -8.71 - 3.02 = -11.73 \\ 8.71 \\ \underline{+ 3.02} \\ 11.73 \end{array}$$

8. What is the difference of $\frac{29}{8}$ and $-2\frac{9}{12}$?

$$\begin{aligned} \frac{29}{8} - \left(-2\frac{9}{12}\right) &= 3\frac{5}{8} + 2\frac{3}{4} = 3\frac{5}{8} + 2\frac{6}{8} = 5\frac{11}{8} \\ &= 6\frac{3}{8} \end{aligned}$$

9. What is the difference of $-5\frac{4}{10}$ and $\frac{36}{7}$?

$$\begin{aligned} -5\frac{4}{10} - \frac{36}{7} &= -5\frac{2}{5} - 5\frac{1}{7} = -5\frac{14}{35} - 5\frac{5}{35} \\ &= -10\frac{19}{35} \end{aligned}$$

10. What is the difference of 8.37 and 12.18 ?

$$\begin{array}{r} 8.37 - 12.18 = -3.81 \\ 12.18 \\ \underline{- 8.37} \\ 3.81 \end{array}$$

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Lesson 3 Practice

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Identify if the product will be positive or negative for questions 1-3.

Expression	Sign of Product
1. $(-3.1)(-2.75)$	positive
2. $\left(\frac{4}{7}\right)\left(-\frac{1}{3}\right)$	negative
3. $(-8.61)(0.45)$	negative

Find the product for questions 4-10.

4. $\left(-3\frac{5}{10}\right)\left(2\frac{3}{4}\right) = -9\frac{5}{8}$

$$-3\frac{1}{2} \cdot 2\frac{3}{4}$$

$$-\frac{7}{2} \cdot \frac{11}{4} = -\frac{77}{8} = -9\frac{5}{8}$$

5. $(-7.2)(-9.15) = 65.88$

$$\begin{array}{r} 9.15 \\ \times 7.2 \\ \hline 1830 \\ 64050 \\ \hline 65880 \end{array}$$

6. $\left(-3\frac{4}{5}\right)\left(2\frac{2}{3}\right) = -10\frac{2}{15}$

$$\frac{19}{5} \cdot \frac{8}{3} = \frac{152}{15} = -10\frac{2}{15}$$

7. $(-5.9)(-6.45) = 38.055$

$$\begin{array}{r} 6.45 \\ \times 5.9 \\ \hline 5805 \\ 32250 \\ \hline 38055 \end{array}$$

$$8. \quad \left(\frac{6}{4}\right)\left(\frac{14}{8}\right) = 2\frac{5}{8}$$

$$\frac{3}{2} \cdot \frac{7}{4} = \frac{21}{8} = 2\frac{5}{8}$$

$$9. \quad (0.56)(-2.15) = -1.204$$

$$\begin{array}{r} 2.15 \\ \times .56 \\ \hline 1290 \\ 10750 \\ \hline 12040 \end{array}$$

$$10. \quad \left(\frac{17}{3}\right)\left(-\frac{20}{6}\right)$$

$$\frac{17}{3} \cdot \frac{-10}{3} = \frac{-170}{9} = -18\frac{8}{9}$$

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Lesson 4 Practice

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Identify if the quotient will be positive or negative for questions 1-2.

Expression	Sign of Quotient
1. $\left(\frac{5}{3}\right) \div \left(-\frac{4}{6}\right)$	negative
2. $(4.5) \div (3)$	positive

Find the quotient for questions 3-10.

3. $(-16.47) \div (-5.4) = 3.05$

$$5.4 \overline{) 16.47} \longrightarrow \begin{array}{r} 3.05 \\ 54 \overline{) 164.70} \\ \underline{-162} \\ 27 \\ \underline{-0} \\ 270 \\ \underline{-270} \\ 0 \end{array}$$

4. $\left(-1\frac{5}{9}\right) \div \left(-\frac{7}{4}\right)$

$$\frac{14}{9} \div \frac{7}{4} = \frac{14}{9} \cdot \frac{4}{7} = \frac{8}{9}$$

5. $(10.32) \div (-0.4) = -25.8$

$$4 \overline{) 103.2} \begin{array}{r} 25.8 \\ 4 \overline{) 103.2} \\ \underline{-8} \\ 23 \\ \underline{-20} \\ 32 \\ \underline{-32} \\ 0 \end{array}$$

6. $\left(\frac{45}{4}\right) \div \left(7\frac{2}{4}\right)$

$$\frac{45}{4} \div \frac{30}{4} = \frac{45}{4} \cdot \frac{4}{30} = \frac{45}{30} = 1\frac{15}{30} = 1\frac{1}{2}$$

$$7. \quad \left(-\frac{7}{5}\right) \div \left(\frac{3}{5}\right) = -\frac{7}{5} \cdot \frac{5}{3} = -\frac{7}{3} = -2\frac{1}{3}$$

$$8. \quad (34.86) \div (4.2) = 8.3$$

$$4.2 \overline{) 34.86} \longrightarrow$$

$$\begin{array}{r} 8.3 \\ 42 \overline{) 348.6} \\ \underline{-336} \\ 126 \\ \underline{-126} \\ 0 \end{array}$$

$$9. \quad \left(-\frac{8}{9}\right) \div \left(\frac{2}{3}\right)$$

$$\frac{-8}{9} \cdot \frac{3}{2} = \frac{-24}{18} = -1\frac{6}{18} = -1\frac{1}{3}$$

$$10. \quad \left(\frac{21}{8}\right) \div \left(-\frac{9}{4}\right)$$

$$\frac{21}{8} \cdot \frac{-4}{9} = \frac{-84}{72} = -1\frac{12}{72} = -1\frac{1}{6}$$

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Lesson 5 Practice

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Use the table below to answer questions 1-3

The table shows transactions for 3 different checking accounts. A negative number represents a debit and a positive number represents a credit.

Account 1	Account 2	Account 3
\$478.92	\$563.25	\$533.30
-\$17.29	-\$62.15	-\$312.55
-\$22.50	\$122.40	\$99.13
\$116.34	-\$49.70	-\$27.15

1. What is the ending balance of Account 1?

$$478.92 - 17.29 - 22.50 + 116.34 = \$555.47$$

2. What is the ending balance of Account 2?

$$563.25 - 62.15 + 122.40 - 49.70 = \$573.80$$

3. What is the ending balance of Account 3?

$$533.30 - 312.55 + 99.13 - 27.15 = \$292.73$$

Use the information below to answer questions 4-5.

Franklin recently went on a weekend hiking trip and recorded the elevation changes at several checkpoints. Franklin's elevation log can be seen in the table below.

Checkpoint	Beginning Elevation (ft)	Final Elevation (ft)
1	153.6	220.8
2	220.8	112.4
3	112.4	88.2
4	88.2	-56.3

Final – Beginning

67.2
-108.4
-24.2
-144.5

4. Which checkpoint has the greatest change in elevation?

Checkpoint 4

5. If the next checkpoint has a final elevation at -87.1, what is the change in elevation?

$$-87.1 - (-56.3) = -30.8 \text{ ft.}$$

6. Jamey purchased a new boat for \$61,014.00. She read that the value of new boats change by $\frac{-2}{19}$ in the first year. What is the value of Jamey's boat after the first year?

$$61,014 \left(\frac{-2}{19} \right) = -6,422.53$$

$$\begin{array}{r} 61014.00 \\ - 6422.53 \\ \hline \$54,591.47 \end{array}$$

7. Jamey learns that the value of her boat will continue to change by $\frac{-3}{10}$ of its value each year after the first year. Using the value after year one from Question 6, what is the value of Jamey's boat at the end of year 2?

$$54,591.47 \left(\frac{-3}{10} \right) = -16,377.44$$

$$\begin{array}{r} 54591.47 \\ - 16377.44 \\ \hline \$38,214.03 \end{array}$$

8. An Orca whale is swimming $-176\frac{1}{4}$ ft below sea level when he notices the rest of his pod above him at the surface. Orca whales can swim at a rate of 120.52 feet per hour. How long, in hours, will it take the Orca to reach his pod?

$$\frac{176.25}{120.52} \approx 1.46 \quad \text{Approximatley 1.46 hours}$$

9. A candle store sells the 3-wick candles for $2\frac{1}{5}$ times what they paid for them from the warehouse. The candles are purchased from the warehouse at \$5.65. How much profit does the candle store make off each candle?

$$2.2(5.65) = 12.43$$

$$\begin{array}{r} 12.43 \\ - 5.65 \\ \hline \$6.78 \end{array}$$

10. After 3 months, the price of the candles is reduced by $\frac{1}{5}$ of the original store price. Using the sale price from Question 9, what is the price of the candles after the 3 month reduction?

$$12.43(0.2) = 2.49$$

$$\begin{array}{r} 12.43 \\ - 2.49 \\ \hline \$9.94 \end{array}$$

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Lesson 6 Practice

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1. Marie's office has a parking garage that charges \$10.50 each day for parking. Marie's employer provides her with a parking card that has \$200 each month. When Marie parked today, the card indicated it had \$116 left on the card. How many days has Marie used the card for parking this month?

$$\frac{(200-116)}{10.50} = 8 \text{ days}$$

2. Tanner is traveling on his first flight to visit family. $8\frac{4}{5}$ minutes into the flight, the pilot announces the plane is currently at an altitude of 3740 feet. At that rate, how long will it take to reach a cruising altitude of 32,400 feet?

$$\frac{8.8}{3740} = \frac{x}{32400} \quad \frac{285,120}{3,740} \approx 76.2 \text{ minutes}$$

3. A penguin sitting on an iceberg notices a fish swimming below. The fish is 10.3 ft below sea level. The penguin dives and reaches the fish in $\frac{7}{15}$ of a minute. What was the rate of descent the penguin made?

$$\frac{10.3 \text{ ft}}{\frac{7}{15} \text{ min}} \approx 22.07 \text{ feet per minute}$$

4. The temperature in North Dakota today was -8.8°C . Over the next 3 weeks, the temperature will increase an average of 3.2°C each week. What will the temperature be at the end of the 3 weeks?

$$3.2(3) = 9.6 \quad -8.8 + 9.6 = 0.8^{\circ}\text{C}$$

5. Lucy buys her cats a bag of food that is 35.3 ounces. Each of Lucy's cats eat $\frac{9}{15}$ ounces a day. If she has 2 cats, how long will the bag of cat food last Lucy?

$$\frac{35.3}{\left(2 \times \frac{9}{15}\right)} = 29\frac{5}{12} \text{ days or approximately 29 days}$$

6. Priscilla is planning to hike a new trail this weekend. She sets up camp Friday evening at the start of the trail which begins at an elevation of -184.56 ft and ends at an elevation of 759.22 ft. She plans to begin her hike in the early morning and wants to travel $\frac{5}{8}$ of the change in elevation of her trip on Saturday, then set up camp until the next morning. What is Priscilla's starting elevation on Sunday?

$$759.22 - (-184.56) = 943.78$$

$$943.78 \cdot \frac{5}{8} = 589.86$$

$$\begin{array}{r} -184.56 \\ + 589.86 \\ \hline 405.3 \text{ ft.} \end{array}$$

7. Kelly is running a race on Monday, which has a total distance of 26.2 miles. Kelly's family wants to show their support by holding up encouraging signs along the path. Her sister decides to stand at the spot that is $\frac{3}{4}$ of the total race while her father chooses to stand at the spot that is $\frac{7}{8}$ of the total race. How many miles apart are Kelly's family members?

$$\begin{array}{ll} \text{Sister} & \text{Father} \\ 26.2 \cdot \frac{3}{4} = 19.65 & 26.2 \cdot \frac{7}{8} = 22.925 \end{array}$$

$$\begin{array}{r} 22.925 \\ - 19.650 \\ \hline 3.275 \text{ mi.} \end{array}$$

8. Kelly runs at a steady rate of 11.2 minutes per mile. Using the information given in Question 7, how many minutes after the race starts should her father expect to see Kelly run by?

$$22.925 \cdot 11.2 = 256.76 \text{ min.}$$

9. Floyd used his store credit card to purchase a new baseball. He has a coupon that when he uses his credit card, the item will be $\frac{2}{5}$ of the total. The baseball was priced at $\$7.25$. A month goes by and Floyd forgot to pay his credit card bill, so they have added a fee of 0.22 times the amount he owed. What is the total amount Floyd has to pay his credit card company?

$$7.25 \cdot \frac{2}{5} = \frac{14.5}{5} = 2.90$$

$$2.90 + 2.90(0.22) = 2.90 + 0.64 = \$3.54$$

10. Stephanie is moving to Miami, Florida from Atlanta, Georgia. The total distance of her move is 665.71 miles. She rented a moving truck that charges $\$22.75$ a day and $\$0.27$ per mile over 500 miles. If she returns the truck with a full tank of gas after 2 days, the company will discount her bill by $\frac{4}{25}$. What is the total amount Stephanie will pay for the moving truck if she fills the tank with gas before returning it after 2 days?

$$\begin{array}{r} 665.71 \\ - 500.00 \\ \hline 165.71 \cdot 0.27 = 44.74 \end{array}$$

$$\begin{array}{r} 22.75(2) = 45.50 \\ + 44.74 \\ \hline 90.24 \cdot \frac{4}{25} = 14.44 \end{array}$$

$$\begin{array}{r} 90.24 \\ - 14.44 \\ \hline \$75.80 \end{array}$$

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Lesson 7 Practice

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Use the scenario for questions 1, 2, and 3.

Valeria and her friends plan to use a rideshare service for their Prom. The service has a booking fee of \$5.50 and charges \$0.86 for the first $1\frac{1}{2}$ mile and \$0.34 for each additional 0.3 mile after the first $1\frac{1}{2}$ mile.

1. How much will Valeria pay if her trip to Prom is 31.5 miles from her home?

$$31.5 - 1.5 = 30 \text{ additional miles} \quad \frac{30 \text{ mi}}{0.3} = 100 \quad 5.50 + 0.86 + (100 \cdot 0.34)$$

$$5.50 + 0.86 + 34$$

$$\$40.36$$

2. Valeria decides to pick up an additional friend. When they arrive at the friend's house, she is not ready to leave. The rideshare service charges \$1.05 for each minute they have to wait. The total bill was \$46.66. How long did Valeria's friend take to get ready?

$$\begin{array}{r} 46.66 \\ -40.36 \\ \hline 6.30 \end{array} \quad \frac{6.30}{1.05} = 6 \text{ minutes}$$

3. Valeria's friends decide to use the same rideshare service to take everyone home. They drop the first friend off at his home 24.6 miles from Prom. The second friend lives another 6.42 miles and Valeria lives another 4.8 miles away. She decides to tip the driver $\frac{3}{8}$ of the bill. How much did Valeria pay for the service, including tip?

$$\begin{array}{r} 24.6 \\ +6.42 \\ \hline 31.02 \\ +4.8 \\ \hline 35.82 \end{array} \quad \begin{array}{r} 35.82 \\ -1.5 \\ \hline 34.32 \end{array} \quad \frac{34.32}{0.3} = 114.4$$

$$5.50 + 0.86 + (114.4 \cdot 0.34)$$

$$5.50 + 0.86 + 38.896$$

$$45.26 \cdot 1\frac{3}{8} \text{ or } 45.26 \cdot 1.375 = \$62.23 \text{ total with tip}$$

4. Diana purchased $2\frac{5}{6}$ pounds of strawberries, $1\frac{2}{3}$ pounds of blueberries, 1.8 pounds of raspberries and 3 oranges to make homemade fruit salad.

Produce	Price (per pound)
Strawberries	\$2.59 per pound
Blueberries	\$2.65 per pound
Orange	\$0.95 each
Raspberries	\$3.09 per pound

Based on the listed prices, how much would Diana spend?

$$\text{Str: } 2\frac{5}{6} \cdot 2.59 = 7.34 \quad \text{BL: } 1\frac{2}{3} \cdot 2.65 = 4.42 \quad \text{R: } 1.8 \cdot 3.09 = 5.56 \quad \text{O: } 3 \cdot 0.95 = 2.85$$

$$7.34 + 4.42 + 5.56 + 2.85 = \$20.17 \text{ total}$$

Use the scenario for questions 5, 6 and 7.

Isabella offers singing lessons for an initial fee of \$10.50 for the first lesson. Each additional lesson she charges \$6.75 for $\frac{1}{2}$ hour and \$4.86 for each additional 0.33 hour after the first $\frac{1}{2}$ hour.

5. Isabella has a new client this week. The first lesson was held on Monday, then an additional lesson on Wednesday. She received payment totaling \$31.83. How many hours was the lesson on Wednesday?

$$\begin{array}{r} 31.83 \\ -10.50 \\ \hline 21.33 \end{array} \quad \begin{array}{r} 21.33 \\ -6.75 \\ \hline 14.58 \end{array} \left(\frac{1}{2} \text{ hour} \right) \quad \frac{14.58}{4.86} = 3 \quad \frac{1}{2} + (3 \cdot 0.33) = 1.49 \text{ or } 1.5 \text{ hours on Wednesday}$$

6. Isabella has a second new client this same week. The first lesson was held on Tuesday, then a second lesson on Thursday for 0.83 hours. On Thursday, how much would she receive for the total singing lessons of this client?

$$\begin{array}{r} 0.83 \\ -0.5 \\ \hline 0.33 \end{array} \quad \begin{array}{r} \text{Half hour} + \text{one } 0.33 \text{ hr} + 1^{\text{st}} \text{ lesson} \\ 6.75 + 4.86 + 10.50 \\ \hline \$22.11 \end{array}$$

7. If Isabella has a third new client this week and only provides the initial singing lesson, how much money in total has she made for the week for all three clients?

$$31.83 + 22.11 + 10.50 = \$64.44$$

8. A cashew store keeps track of the weekly changes in their inventory. Cashews are delivered to the shop in $20\frac{5}{6}$ -pound (lb) bags, 18 bags at a time twice a week. The inventory at the end of week shows a total of $177\frac{3}{4}$ lbs. If the same amount of cashews were sold each day, how many pounds of cashews did they sell in one day?

$$18 \cdot 2 = 36 \text{ total bags} \quad \begin{array}{r} 750 \\ -177.75 \\ \hline 572.25 \text{ lbs. sold} \end{array} \quad \frac{572.25}{7} = 81.75 \text{ lbs per day}$$

$$36 \cdot 20\frac{5}{6} \text{ lbs} = 750 \text{ total lbs.}$$

9. Devon is building birdhouses for the neighborhood birds. To make one birdhouse, he needs 4 pieces of wood $\frac{3}{4}$ feet for the walls, 1 piece of wood 0.25 feet for the floor and 2 pieces of wood $\frac{11}{24}$ feet for the roof. A single board is 8.5 feet long and costs \$10.28. If Devon has \$35, how many birdhouses can he make?

$$4(0.75) + 0.25 + 2\left(\frac{11}{24}\right) = 3 + 0.25 + \frac{11}{12} = 4\frac{1}{6} \text{ ft./birdhouse} \quad \frac{35}{10.28} = 3 \text{ boards she can buy}$$

$$8\frac{1}{2} \div 4\frac{1}{6} = 2\frac{1}{25} = \text{birdhouses per board} \quad 3 \cdot 2 = 6 \text{ birdhouses}$$

10. Bailey is making bows to sell to support her cheerleading team. She purchased a roll of ribbon that is 53.17 ft in length and each bow uses $3\frac{1}{8}$ feet to make. If she makes 10 bows, how many more bows can be made with the ribbon left in the roll?

$$10 \cdot 3\frac{1}{8} = 31\frac{1}{4} \text{ ft used} \quad 21.92 \div 3\frac{1}{8} = 7$$

$$53.17 - 31.25 = 21.92 \text{ ft. left} \quad 7 \text{ more bows.}$$

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Lesson 8 Practice

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1. Joshua says that when he evaluated an expression, the result was 100. Use mental math to estimate each value. Then, circle which of the following expressions could have been Joshua's expression.

$$(10 + 2)^2 - 44$$

$$10^2 + 2^2 - 44$$

$$(10 - (-2)^2) + 44$$

$$12^2 + 44$$

2. Cindy evaluated the expression $3^4 \div 3(2.5 + 5.5)$.

Circle Cindy's error and choose which step of the order of operations she did not apply correctly.

Cindy's Steps	Order of Operations
$3^4 \div 3(2.5 + 5.5)$ $3^4 \div 3(8)$ $81 \div 3(8)$ $81 \div 24$ 3.375	• Evaluate expressions with exponents • Perform operations inside grouping symbols. • Perform multiplication and division in order from left to right. • Perform addition and subtraction in order from left to right.

3. Evaluate the expression without using a calculator. Write an explanation using the order of operations next to each step.

$ 7 - 10 \div (-6)$	Explanation
$ -3 \div (-6)$	grouping symbols
$-3 \div (-6)$	grouping symbols
$\frac{-3}{-6} \div 3 = -\frac{1}{2}$	Divide from left to right

4. Evaluate the expression without using a calculator. Write an explanation using the order of operations next to each step.

$(4 \times 1.5)^2 \div 9$	Explanation
$(6)^2 \div 9$	grouping symbols
$36 \div 9$	Evaluate expressions with exponents
4	Divide from left to right

Evaluate each expression.

5. $\frac{(-12 \times 0.25)^3}{9} = \frac{(-3)^3}{9} = \frac{-27}{9} = -3$

6. $5 \times |-8.8 \times 2.2| = 5 \times |-19.36| = 5 \times 19.36 = 96.8$

7. $(-92 \times 0.3)^2 + 52.24 = (-27.6)^2 + 52.24 = 761.76 + 52.24 = 814$

Determine if the equations in the table are true or false. Show work to support your answer.

Equation	True or False?
8. $3 - 3 \times 3^3 = -27$	$3 - 3 \times 27$ $3 - 81$ -78 True
9. $ 9.2 - 6.7 \times (-2) = -5$	$ -2.5 \times (-2)$ $2.5 \times (-2)$ -5 True
10. $5.1 \times 8.5 + (-4)^2 = 39.35$	$5.1 \times 8.5 + 16$ $43.35 + 16$ 59.35 False

Grade 6 Accelerated
Unit 11
Lesson 9 Practice

Name Answer Key
Date _____
Period _____

In the table an expression has been evaluated, but the steps are mixed up. Order the expression using the numbers 1-5 on the left, the final answer has been given to you. Write an explanation for each step.

Step	Expression	Explain the Step
5	$150 - (17.64)$	Exponent
1	$ -9.392 + (-5.2)^3 - (-4.2)^2$	Original expression
4	$150 - (-4.2)^2$	Parenthesis
2	$ -9.392 + (-140.608) - (-4.2)^2$	Exponent within parenthesis
3	$ -150 - (-4.2)^2$	Simplify within parenthesis
Answer	132.36	Subtract

Evaluate each expression without using a calculator. Show work to support your answer.

6. $\left(\frac{4}{5}\right)^2 \times 25 + 4^2$

$$\frac{16}{25} \times 25 + 4^2$$

$$\frac{16}{25} \times 25 + 16$$

$$16 + 16$$

$$32$$

$$7. \quad [(11 - 3.5)^2 - 6.25]^2 \div 50$$

$$\begin{aligned} & [7.5^2 - 6.25]^2 \div 50 \\ & [56.25 - 6.25]^2 \div 50 \\ & 50^2 \div 50 \\ & 2500 \div 50 \\ & 50 \end{aligned}$$

$$8. \quad 12.2 - (3^3 - 2)^2 \div 25$$

$$\begin{aligned} & 12.2 - (27 - 2)^2 \div 25 \\ & 12.2 - 25^2 \div 25 \\ & 12.2 - 625 \div 25 \\ & 12.2 - 25 \\ & -12.8 \end{aligned}$$

$$9. \quad -\left|4 - \frac{3}{4}\right| + \frac{3}{10}(3^2 - 49)$$

$$-\left|3\frac{1}{4}\right| + \frac{3}{10}(9 - 49)$$

$$-3\frac{1}{4} + \frac{3}{10}(-40)$$

$$-3\frac{1}{4} - 12$$

$$-15\frac{1}{4}$$

$$10. \quad \frac{(-8+|-5.5-6.5|)^3}{16}$$

$$\frac{(-8+|-12|)^3}{16}$$

$$\frac{(-8+12)^3}{16}$$

$$\frac{4^3}{16}$$

$$\frac{64}{16}$$

$$4$$